PRAMUK WIJERATHNE

AI Solutions Architect | Agentic AI & RAG | Azure | Hands-on Delivery

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NZ Permanent Resident | Available at short notice | Permanent or contract | Onsite, hybrid or remote

PROFILE

AI solutions architect with 22+ years of enterprise software engineering, who designs AI systems and then builds them. At Inov8 I took two generative AI systems for New Zealand public-sector clients from architecture through to deployment single-handedly; one has now been live in production at New Zealand Transport Agency for close to a year. I bring solution design, Python engineering, retrieval and model choice, evaluation, security and Azure deployment in one person, and I leave teams with documented systems they can run and extend.

CAREER HIGHLIGHTS

- **AI shipped to production, not just advised on:** Architected and built CAS-RAG for New Zealand Transport Agency, live in production since late 2025: a GPT-4.1 tool-calling agent inside a deterministic orchestrator, classifying free-text crash narratives against a 345-code taxonomy across 177K+ cases. Roughly a 50% relative improvement over the prior retrieval design and about 2.4x a single-shot LLM baseline.
- **End-to-end ownership:** Designed, built and deployed CAS-RAG and GovChat solo through the full SDLC, using a team of AI QA agents alongside client UAT. Clients get architect-level decisions and engineer-level output from one person.
- **Cost-aware engineering:** A Redis semantic cache cut inference cost by 40 to 60%, and a fine-tuned, INT8-quantised DeBERTa classifier runs at about 25ms per case on CPU, served over MCP so any agent can reuse it and sensitive data stays inside the client boundary.
- **Production-grade AI governance:** Evaluation harnesses, shadow A/B testing against deployed endpoints, drift heartbeats, deterministic validators, human-in-the-loop review, RBAC/JWT and OpenTelemetry tracing. Changes ship on evidence, and every AI decision is auditable.
- **Fast, credible discovery and PoCs:** Reference architectures, security frameworks, risk assessments and roadmaps for New Zealand government RFPs, RFIs and RFQs, with working proofs of concept built during procurement.
- **Government and regulated-sector track record:** Technical lead on Ministry of Justice and DIA work at Datacom; solution architect for Singapore government at NCS; payment platforms for ten banks across Asia Pacific at BPC. Winner, GovHack 2025.

TECHNICAL SKILLS

Agentic AI & Orchestration	LangChain, LangGraph, LangChain Deep Agents, MCP (client and server), tool-calling agents, agentic RAG, Graph RAG (Neo4j), Microsoft Agent Framework, planner / executor / validator patterns, human-in-the-loop
LLMs & Models	Azure OpenAI (GPT-4.1 family), embeddings, DeBERTa fine-tuning, INT8 quantisation, Hugging Face, TensorFlow, prompt optimisation
Retrieval & Knowledge	pgvector (HNSW), Azure AI Search, Redis semantic cache, Pinecone, Chroma, Qdrant, document ingestion (Tesseract OCR, Azure Document Intelligence), chunking and re-ranking
Evaluation & Governance	Offline and shadow A/B evaluation, drift monitoring, deterministic validators, content moderation, OpenTelemetry / Jaeger, RBAC, JWT, Entra ID, OAuth 2.0, OWASP Top 10
Architecture & Advisory	Solution and reference architecture, security frameworks, risk assessments, delivery roadmaps, RFP / RFI / RFQ responses, proofs of concept, stakeholder engagement, leading teams of up to 12
Languages & Frameworks	Python (primary for AI work), FastAPI, C# / .NET 8, Java / Spring Boot, C, React, JavaScript; TypeScript (working knowledge)
Cloud & Platform	Azure (OpenAI, AI Search, Container Apps, Functions, API Management, Key Vault, Service Bus), Docker, Kubernetes (AKS, EKS), Bicep, Terraform; AWS in addition to Azure
CI/CD & Tooling	Azure DevOps, GitHub Actions, Bamboo, Jenkins, Git, SonarQube; AI-assisted engineering with Claude Code, Cursor and GitHub Copilot
Integration & Data	REST / OpenAPI, event-driven microservices, IBM MQ, RabbitMQ, WSO2; PostgreSQL, Oracle, SQL Server, Cosmos DB, MongoDB

WORK EXPERIENCE

AI Solutions Architect
Jul 2025 – Present

Inov8 Limited | Wellington, New Zealand

- Designed and built CAS-RAG end to end for New Zealand Transport Agency: label-aware pgvector retrieval, a DeBERTa classifier whose probabilities feed the agent loop as priors the LLM can confirm or override, and a deterministic validator layer before any result is committed. Live in production for close to a year after UAT and security review.
- Exposed the prediction pipeline as MCP tools, turning a single application into a reusable service other agents and workflows can call.
- Built the evaluation and monitoring layer: per-case scoring against deployed endpoints, shadow A/B runs for model selection, and a scheduled heartbeat that re-scores a fixed holdout to catch silent regressions.
- Designed a human-in-the-loop correction loop where expert corrections, gated by multi-reviewer consensus, flow back into the retrieval index without retraining.
- Built GovChat, a resident assistant for a New Zealand district council, against a live RFQ: hybrid intent router, RAG agent with re-ranking and query expansion, planner / executor / validator pipeline, content moderation, OCR document ingestion, RBAC/JWT and OpenTelemetry tracing. Deployed to Azure Container Apps and awaiting council security review.
- Now leading new classification proofs of concept, reusing the CAS-RAG patterns, and running the technical side of New Zealand government RFP, RFI and RFQ responses.

Technologies: Python, FastAPI, LangChain, LangGraph, MCP, Azure OpenAI, Azure AI Search, pgvector, Redis, DeBERTa, Container Apps, Key Vault, Docker, Kubernetes, Bicep, OpenTelemetry, React.

Senior Software Developer (Technical Lead)
Jan 2023 – May 2025

Datacom | Wellington, New Zealand

- Owned CI/CD automation, release support and database performance tuning for the Ministry of Justice HUB system, working with Fujitsu and MoJ teams (team of 8).
- Led development of the UCIC and Court Management System (Java, EJB, WebLogic, Oracle) for a team of 12, including NZTA safety camera infringement integration and Firearms and Gangs Act legislative changes.
- Built secure, API-driven services on Azure API Management, Azure Functions and Entra B2B; delivered work for DIA, Dell, Downer, DairyNZ and Spark, plus Power Platform and SharePoint automation.

Technologies: C# / .NET 8, Java, Spring Boot, Oracle 21c, IBM MQ, Azure DevOps, Bamboo, Bitbucket, Azure API Management, Azure Functions, Docker, AKS, EKS.

Solutions Architect (Lead Service Delivery Manager)
Oct 2021 – Oct 2022

NCS Pte Ltd (Singtel Group) | Singapore

- Led cross-functional delivery teams (developers, QA, BA) for Singapore government agencies; owned architecture and detailed design, and resolved complex issues hands-on.
- Delivered an EJB to Struts2 migration milestone as part of an application modernisation programme, and managed SLA, finance and ITIL alignment with the programme manager.

Product Development Manager
Oct 2019 – Sep 2021

Assimilated Technologies Pte Ltd | Singapore | Team of 6

- Re-architected a monolithic access and visitor management platform into event-driven microservices on Azure Functions and Service Bus, for clients including Resorts World Sentosa, ICA Singapore, banks and airports.
- Won the Resorts World Sentosa guest check-in bid by building the MVP, then delivered the production platform, cutting check-in time from about 3 minutes to about 45 seconds.
- Built secure OAuth 2.0 / OpenAPI integrations with Opera, DBS POS and Alipay, plus OCR and liveness detection for kiosks.

Technical Lead Developer
Apr 2019 – Aug 2019

Allied Container Group | Singapore | Team of 4

- Broke legacy monolithic services into microservices and integrated them with third-party systems.

Technical & Implementation Lead (Payment Expert)
Oct 2014 – Mar 2019

BPC Banking Technologies | Singapore | Team of 5 to 8

- Led SmartVista payment switching deployments on Unix / Oracle for ten banks and payment institutions across Sri Lanka, Nepal, Indonesia, New Zealand, Myanmar, the Philippines and Cambodia.
- Integrated ATM, POS, kiosk, mobile and internet banking channels over ISO 8583; certified on Visa, Mastercard and UPI; worked with HSMs (Thales, SafeNet) and core banking systems (Finacle, Flexcube).

Earlier Career 2004 – 2014

- Associate Technical Lead, hSenid Software International (Sri Lanka): NoSQL reporting redesign and WSO2 middleware stack.
- Senior Software Engineer, Codegen International (London / Sri Lanka): TravelBox booking engines for travel clients in the UK, Canada and the Netherlands.
- Software Engineer, Wavenet International, Lanka Bell and Celetronix (Sri Lanka): SMSC, IVR, SIP and VoIP platforms, telecom billing, and low-level hardware test software.

EDUCATION

BSc (Special) Engineering, Computer Engineering 1999 – 2004
University of Peradeniya, Sri Lanka | Merit Pass | GPA 3.45/4.2

Postgraduate Diploma in Business Administration (IT) 2007 – 2009
University of Moratuwa, Sri Lanka

AWARDS & CERTIFICATIONS

- **Loop Engineering with LangChain Deep Agents** | *DeepLearning.AI* Oct 2026
- **Agent Skills with Anthropic** | *DeepLearning.AI* Feb 2026
- **Winner, Unlocking Productivity Through Connected Health Data, GovHack 2025** | *Infosys* Dec 2025
- **Agentic AI Certification** | *DeepLearning.AI* Oct 2025
- **Building Live Voice Agents with Google's ADK** | *DeepLearning.AI* Oct 2025
- **Build AI Apps with MCP Server: Working with Box Files** | *DeepLearning.AI* Sep 2025
- **Microsoft Certified: Azure Fundamentals (AZ-900)** | *Microsoft* Feb 2023

PUBLICATIONS & AFFILIATIONS

- Label-Aware RAG: Asymmetric Indexing for Multi-Label Classification at Scale (Medium)
- Why I Fine-Tuned DeBERTa Instead of Asking the LLM Harder (Medium)
- **Affiliations:** Member, IEEE; Associate Member, Institution of Engineers Sri Lanka (IESL); Engineering Council Sri Lanka (ECSL).